



Investigating the effect of Input Modality on Reasoning in VLMs using Zero-shot evaluation

Rodela Ghosh¹ Aviral Gupta¹

¹Bellini College of AI, Cybersecurity and Computing, University of South Florida

Abstract

This project investigates how input modality affects mathematical reasoning in vision-language models (VLMs). Using 100 samples from the GSM8K benchmark [1], we evaluate Qwen2-VL-2B-Instruct [2] and LLaVA-v1.6-Mistral-7B [3] across three conditions: text-only, rendered image, and a modality mismatch condition where image and text describe different problems. Both models show substantial accuracy drops under visual input — Qwen2 falls from 55% to 30% and LLaVA from 40% to 21%. In the mismatch condition, both models follow the text modality in over 75% of cases, revealing strong text dominance when modalities conflict. These findings suggest that rendering math problems as images meaningfully impairs reasoning in current VLMs, even when visual content is clean and legible

General Project Description and Goals

Vision-language models (VLMs) increasingly handle tasks combining visual and textual input, yet their behavior under conflicting modalities remains poorly understood. Mathematical reasoning offers a controlled testbed — problems have unambiguous answers and identical content can be presented across modalities without changing meaning.

Goal 1 — Text-only baseline. Evaluate Qwen2-VL-2B and LLaVA-1.6-7B on raw GSM8K text to establish a language-only reference point.

Goal 2 — Visual input impact. Present the same problems as clean rendered PNG images to directly measure whether vision helps, hurts, or is neutral to reasoning accuracy.

Goal 3 — Modality conflict. In a novel mismatch condition, models receive the image of one problem and the text of another. Numeric distance scoring reveals which modality the model implicitly trusts. Findings inform VLM deployment in educational tools, assistive technologies, and document understanding systems where math frequently appears in visual formats.

GSM8K Zero-Shot Accuracy by Input Modality

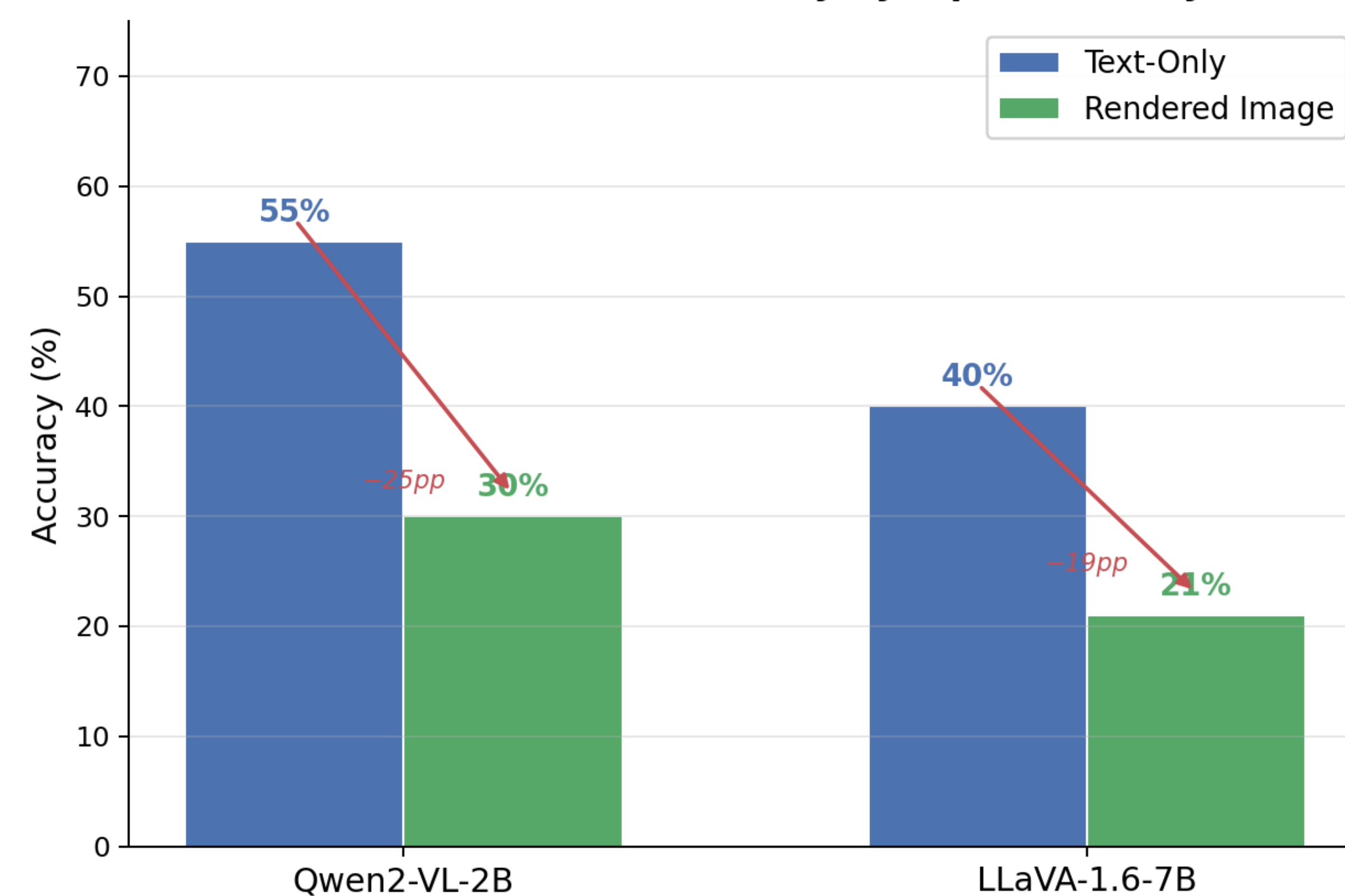


Figure 1. Zero-shot accuracy on GSM8K across input modalities (n=100)

Experiments

Objective: Evaluate how input modality affects mathematical reasoning accuracy across two VLMs using the GSM8K benchmark.

Dataset: GSM8K test split — first 100 problems used for evaluation.

Models: Qwen2-VL-2B-Instruct, LLaVA-v1.6-Mistral-7B

Input Conditions:

Condition 1 — Text-Only: Raw problem text passed directly to the model; vision encoder unused

Condition 2 — Rendered Image: Problem text converted to a clean PNG image via Pillow; vision encoder active

Condition 3 — Modality Mismatch: Model receives image of problem i alongside text of problem $i+1$; tests implicit modality preference

Independent Variable: Input modality (text, rendered image, mismatch)

Dependent Measures: Top-1 answer accuracy, error type distribution (arithmetic, reasoning, no number), numeric distance to each modality's ground truth answer (Condition 3)

Procedure: Load GSM8K samples → render into target format → run inference → extract final numeric answer → compare to ground truth → classify errors

Controls: Fixed 100-problem subset, fixed answer extraction pipeline, greedy decoding (do_sample=False), Google Colab T4 GPU

Results

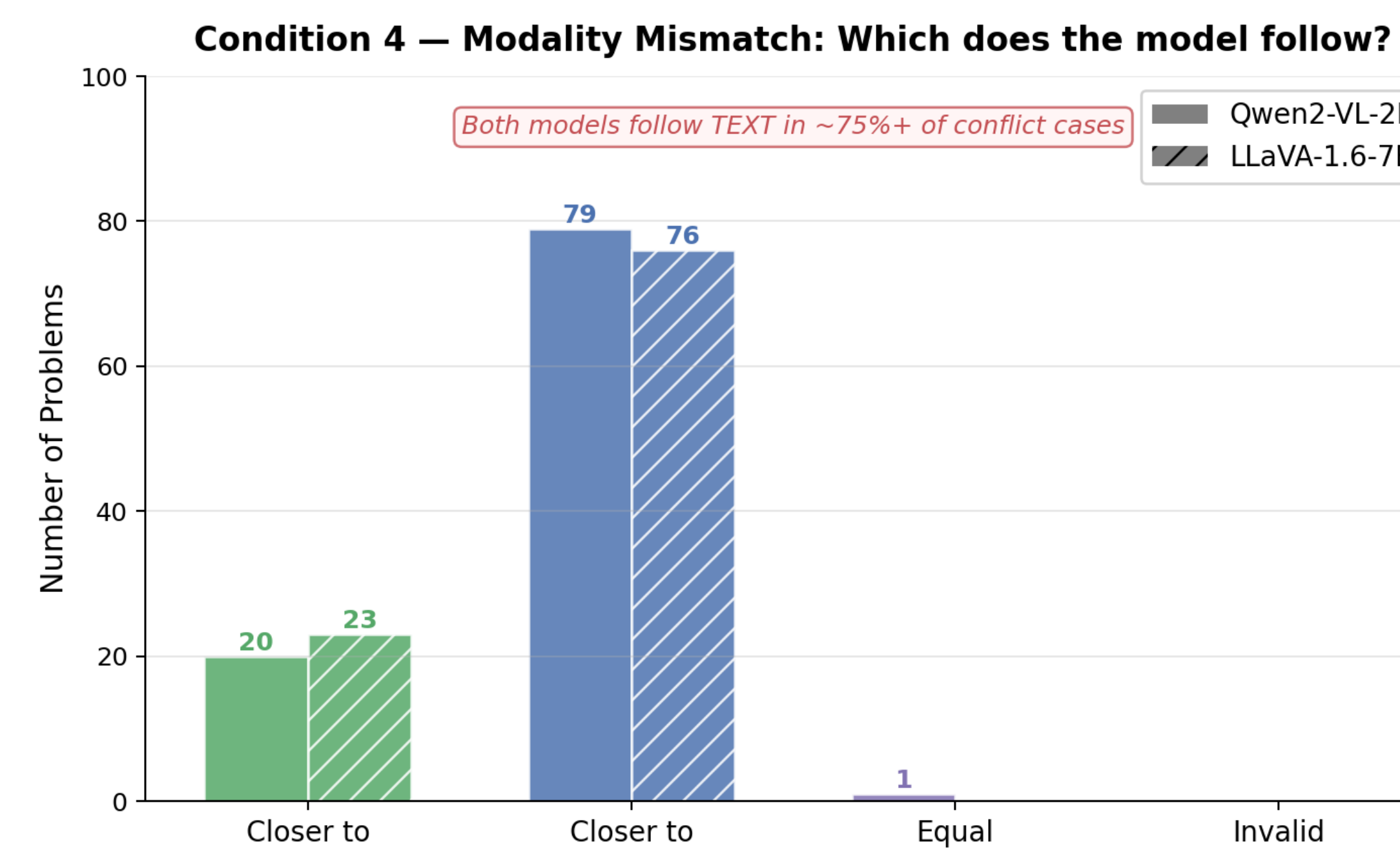


Figure 2: Modality preference under conflict — numeric distance scoring (n=100)

Summary of Experiments



Figure 3: Error type distribution by condition and model

Discussion

Results reveal a consistent pattern across both architectures: presenting math problems visually causes substantial accuracy drops: Qwen2-VL-2B falls 25 percentage points (55% → 30%) and LLaVA-1.6-7B falls 19 points (40% → 21%). The consistency across two different model families suggests this reflects a fundamental limitation in how VLMs process visually encoded symbolic text, rather than a model-specific weakness.

Error analysis identifies arithmetic errors as the dominant failure mode under visual input, rising from 45 → 68 for Qwen2 and 60 → 77 for LLaVA. Critically, vision/OCR errors remain near zero for both models, indicating the models can read rendered text but lose computational precision in doing so. Visual encoding appears to disrupt arithmetic reasoning without impairing text recognition.

The mismatch condition provides the most diagnostic finding: when image and text describe different problems, both models overwhelmingly follow the text — Qwen2 in 79/100 cases, LLaVA in 76/100. This reveals a strong implicit text bias in multimodal integration. The vision encoder contributes minimally when competing text is present, which likely explains why visual input degrades rather than augments performance — the model attends to text while the vision pathway introduces noise.

Future Work

- Scale evaluation to the full GSM8K test set to strengthen statistical reliability
- Add significance testing to formally validate accuracy differences and mismatch dominance
- Implement screenshot condition with controlled noise levels to test real-world robustness
- Evaluate frontier models (GPT-4o, Gemini 2.0) to assess whether text dominance persists at scale
- Test on additional math benchmarks such as SVAMP and AQUA-RAT to assess generalizability
- Optimize inference pipeline for low-resource GPU environments

Acknowledgments

This project was conducted through the PALM Lab at the University of South Florida under the UR2PhD program. The authors thank graduate mentors Ocean Monjur and Shrestha Datta for their guidance and mentorship throughout this research. We also thank PI Anshuman Chhabra and the UR2PhD team for providing research training, resources, and the opportunity to present this work.

References

- [1] Cobbe et al. (2021). Training Verifiers to Solve Math Word Problems. arXiv:2110.14168.
- [2] Wang et al. (2024). Qwen2-VL: Enhancing Vision-Language Model's Perception of the World at Any Resolution. arXiv:2409.12191.
- [3] Liu et al. (2024). LLaVA-NeXT: Improved Baselines with Visual Instruction Tuning. arXiv:2310.03744.